Task #322 v0.8 — Multi-phase quiver explicit kQ path algebra framework

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A_sub v0.8 — Multi-phase quiver explicit kQ framework

1. Cal #29 STANDING question-shape audit (applied at design)

Question: "What is the explicit path algebra kQ for the 36-node Phase A v0.2 multi-phase quiver Q, with arrow set from 10 Cal-verified A_sub commutators and σ_BF Z_2 grading?"

Audit: - Structurally determined? YES — standard quiver representation theory machinery (Gabriel 1972 + path algebra construction) - Back-fittable? NO — kQ is determined by quiver structure + relations - Pre-suppositions? Cal #132 SVC for 8 commutators + step 10 SVC CANDIDATE + σ_BF/γ^5 disambiguation (all explicit)

Pass: question shape is structural. Cal #29 STANDING discipline holds.

2. Quiver Q = (V, A) explicit definition

2.1 Vertex set V

V = 36 Wallach K-types from Elie Toy 3537 Phase A v0.2 (cutoff m_1 + m_2 ≤ 7):

V = {V_(m_1, m_2) : m_1 ∈ {0, 1/2, 1, 3/2, 2, 5/2, 3, 7/2, 4, 9/2, 5, 11/2, 6, 13/2, 7}, m_2 ∈ {0, 1/2, 1, 3/2, …, 7/2}, m_1 ≥ m_2, m_1 + m_2 ≤ 7, σ_BF consistency}

Vertex partition by σ_BF: - |V_boson| = 20 (integer m_1 + integer m_2) - |V_fermion| = 16 (half-integer m_1 + half-integer m_2) - |V_mixed-forbidden| = 0 (Pin(2) Z_2 grading consistency)

Total: |V| = 36 vertices.

2.2 Arrow set A — by A_sub generator class

Per v0.7 Section 3 + step 10 closure, A_sub's 15 generators (14 from v0.1 + σ_BF explicit) induce arrows:

Class 1 — Diagonal arrows (self-loops at each vertex): - Ĥ_sub (Casimir): self-loop with weight C_2(K) - N̂ (K-type level): self-loop with weight m_1 + m_2 - σ_BF (Pin(2) Z_2 grading): self-loop with weight ε_K ∈ {+1, -1} - $\hat{\gamma}^5$ (Dirac chirality): self-loop on fermion vertices only, weight χ_K ∈ {+1, -1}; UNDEFINED on bosons - Q̂ (charge): self-loop with weight m_2 - Î_3 (weak isospin Cartan): self-loop with weight related to m_2

Self-loop count: 6 generators × 36 vertices = 216 self-loop arrows (with $\hat{\gamma}^5$ undefined on 20 boson vertices → reduces to 196 effective).

Class 2 — Within-sublattice m_1-direction arrows: - x̂*i (Hua coordinates, i = 1..n_C = 5): arrows V*(m_1, m_2) → V_(m_1 + 1, m_2) and V_(m_1, m_2) → V_(m_1 - 1, m_2) - p̂_i (Wirtinger derivative, i = 1..5): same arrows (commute with x̂_i to give canonical relation) - L̂_i (so(5) generators, 10 total): act within SO(5) multiplet at fixed (m_1, m_2)

Within-sublattice arrow count for 36-node subgraph (manual count): - Boson m_1-increase arrows: ~15 (e.g., (0,0)→(1,0), (1,0)→(2,0), …, (6,0)→(7,0); (1,1)→(2,1); etc.) - Fermion m_1-increase arrows: ~12 (e.g., (1/2,1/2)→(3/2,1/2), (3/2,1/2)→(5/2,1/2), etc.) - Each generator x̂_i, p̂_i contributes parallel arrows → 5 + 5 = 10 generators × ~27 m_1-increase arrows = ~270 arrows (with corresponding decrease arrows by hermitian conjugation, ~540 arrows total)

Class 3 — Cross-sublattice arrows: - Per step 10 closure: none from Ŝ_i (spin algebra doesn't connect sublattices; no SUSY) - Per discrete symmetries T̂, Ĉ, P̂_op: same sublattice (preserve particle type via σ_BF commutation) - Per Möbius involution σ (in P̂_op): m_2 → -m_2 with phase; for 36-node set with m_2 ≥ 0 (Wallach holomorphic discrete series convention), σ acts trivially at m_2 = 0 nodes; exits Wallach range elsewhere

Cross-sublattice arrow count: structurally 0 (no boson↔fermion arrows in pure A_sub).

Class 4 — Spin-fiber arrows (intra-vertex): - Per step 10 closure: Ŝ_i edges live in spin-fiber over each K-type node, not as inter-K-type arrows - 3 spin generators × 36 vertices give intra-vertex arrows; fiber structure - Same for color SU(3) Ĉ_3 (8 generators × 36 vertices) and weak isospin Î_± (raising/lowering, 2 generators × 36 vertices)

Fiber arrow count: (3 + 8 + 2) × 36 = 468 intra-vertex arrows; treated as FIBER STRUCTURE not main-graph arrows.

Class 5 — Bell-CHSH arrow: - B̂ rank-1 projector: single arrow V_(0,0) → V_(0,0) self-loop with weight β (per T2399) - [B̂, T̂_tick] rank-1: arrow V_(0,0) → V_(1,0) with weight β · (step 9 FRAMEWORK-PLUS)

Bell arrow count: 2 arrows (1 self-loop + 1 vacuum kicker).

2.3 Total arrow count estimate

Main quiver (excluding fiber): - Self-loops: 196 - Within-sublattice m_1: ~540 - Cross-sublattice: 0 - Bell: 2 - Substrate-tick T̂_tick (model-dependent): ~36 (one per node, advancing m_2 or m_1 per simplest model)

Total main-graph arrows: ~774 arrows for Phase A v0.2 36-node set.

Fiber structure: 468 intra-vertex arrows (color/spin/weak isospin internal multiplet structure).

Total quiver structure: 36 vertices + ~774 main arrows + 468 fiber arrows = ~1278 directed-graph elements for Phase A v0.2.

3. Path algebra kQ definition

3.1 Field k

For BST substrate on Bergman $H^2(D\_IV^5)$, the natural base field is k = ℂ (complex Bergman Hilbert space). All matrix elements are complex; representations are complex vector spaces.

3.2 Path algebra kQ construction

Definition: kQ is the path algebra of quiver Q over field k. Elements are formal sums of PATHS in Q. A path p of length n is a sequence of composable arrows: p = α_n α_{n-1} … α_1 with s(α_{k+1}) = t(α_k) (each arrow ends where the next begins).

Basis of kQ: - Trivial paths (length 0): one per vertex; act as identity at each vertex - Length-1 paths: single arrows - Length-n paths: composed arrows

Multiplication: path concatenation when composable; zero otherwise.

Dimension (for finite Phase A cutoff): finite when all paths within 36-node set are bounded by cutoff; possibly INFINITE for unbounded path lengths.

3.3 Phase A finite cutoff

For the Phase A v0.2 36-node cutoff (m_1 + m_2 ≤ 7), paths within the 36-node set: - Bounded by Casimir cutoff (paths can't exit the 36-node set; otherwise undefined) - Each x̂_i raise arrow increases m_1 by 1; finite number of consecutive m_1-raises before exit - Maximum m_1-raise path length ~7 (from (0,0) to (7,0)) - Maximum m_2-raise path length ~3-4 (constrained by m_1 ≥ m_2)

Finite-dim kQ_Phase A: approximately bounded by (#arrows)^max_path_length ≈ 774^7 = ~10^20 dim. TOO LARGE for direct computation without relations applied.

Strategy: apply commutator relations IMMEDIATELY to reduce kQ_Phase A to its quotient kQ_Phase A / ⟨R⟩ where R is the relation ideal.

3.4 Relation ideal R from A_sub commutators

The 10 closed A_sub commutators (8 SVC + step 9 FRAMEWORK-PLUS + step 10 SVC CANDIDATE) generate the relation ideal R. Each commutator [X_i, X_j] = Σ_k c^k_ij X_k (or = 0 for vanishing commutators) gives a relation on path products:

Universal relations (active in all phase regions): 1. $\{\hat{Q}, \hat{P}\text{\_op}\} = 0 \to \hat{Q}\hat{P}\text{\_op} + \hat{P}\text{\_op}\hat{Q} = 0$ as path equality 2. $\{\hat{\gamma}^5, \hat{T}\} = 0 \to \hat{\gamma}^5\hat{T} + \hat{T}\hat{\gamma}^5 = 0$ 3. $\{\hat{\gamma}^5, \hat{C}\} = 0 \to \hat{\gamma}^5\hat{C} + \hat{C}\hat{\gamma}^5 = 0$ 4. $[\hat{\gamma}^5, \hat{P}\text{\_op}] = 0 \to \hat{\gamma}^5\hat{P}\text{\_op} - \hat{P}\text{\_op}\hat{\gamma}^5 = 0$ 5. $[\hat{B}, \hat{Q}] = 0 \to \hat{B}\hat{Q} - \hat{Q}\hat{B} = 0$ 6. $[\hat{L}\text{\_i}, \hat{\gamma}^5] = 0 \to \hat{L}\text{\_i}\hat{\gamma}^5 - \hat{\gamma}^5\hat{L}\text{\_i} = 0$ 7. $[\hat{C}\text{\_3}, \hat{I}\text{\_3}] = 0 \to \hat{C}\text{\_3}\hat{I}\text{\_3} - \hat{I}\text{\_3}\hat{C}\text{\_3} = 0$ 8. $[\hat{S}\text{\_i}, \hat{S}\text{\_j}] = i\hbar\varepsilon\text{\_ijk}\ \hat{S}\text{\_k}$ (SU(2) algebra; structural)

Region-dependent relations (active in specific phase regions per v0.5): - DIRECT: all 10 commutators active - COMPOSITION: ABJ anomaly relations + RG-flow relations (multi-week derivation) - MATERIAL: external BC relations (multi-week) - COMBINATORIAL: gauge-dimension identities + Mersenne ladder + Bergman 225 (multi-week)

Multi-phase relation ideal: R = R_DIRECT ? R_COMPOSITION ? R_MATERIAL ? R_COMBINATORIAL (different relations active in different sub-quivers).

3.5 The quotient algebra kQ / ⟨R⟩

The substrate's path algebra is kQ / ⟨R⟩ where ⟨R⟩ is the ideal generated by R. This produces a quotient with: - Path equivalence classes (paths equal modulo R) - Reduced dimension (after relation application) - Substantive algebraic structure encoding A_sub commutator content

Estimate for Phase A 36-node finite-dim quotient: after relation application, dimension reduces from ~10^20 to manageable size. Precise count is v0.3+ multi-week computation work.

4. Representations of (Q, R) = kQ-modules

A representation of (Q, R) assigns: - Vector space V_x to each vertex x ∈ V - Linear map ρ_α : V_{s(α)} → V_{t(α)} to each arrow α ∈ A - Satisfying relations R: ρ-image of each relation evaluates to zero in End(⊕ V_x)

For BST substrate: - V_x = (Wallach K-type V_x as ℂ-vector-space) for each K-type vertex - ρ_α = (matrix element of A_sub generator action) for each arrow - Relations R = (commutator-induced path equalities) hold automatically because they come from A_sub's algebra

The substrate's actual H^2(D_IV^5) IS a representation of (Q, R) with: - V_x = irreducible K-type representation of K = SO(5) × SO(2) (or Spin(5) × Pin(2) cover on fermion sublattice) - ρ_α = explicit matrix element computation per Wallach formulas

This is the canonical representation of the substrate quiver.

5. Multi-phase structure — 4 sub-quivers

Per v0.5 + v0.6: the substrate quiver Q decomposes into 4 sub-quivers sharing nodes:

Q = Q_DIRECT ⊕ Q_COMPOSITION ⊕ Q_MATERIAL ⊕ Q_COMBINATORIAL

(Direct sum at the level of arrow sets; nodes shared universally.)

Sub-quiver	Active arrows	Phase region
Q_DIRECT	Steps 1-9 + step 10 + universal Class 1	57% catalog observables
Q_COMPOSITION	Steps 1-5, 8, 9 + ABJ anomaly + RG-flow	10%+
Q_MATERIAL	Steps 7, 8 + external BC	20%
Q_COMBINATORIAL	Steps 3-5, 8 + Mersenne + gauge-dim + 225	13%

Each sub-quiver has its own path algebra kQ_phase / ⟨R_phase⟩ with phase-specific relations. The full substrate quiver Q is the union with the combined relation ideal.

[Ĉ_3, Î_3] = 0 universally: active in ALL 4 sub-quivers (gauge factorization preserved everywhere).

6. Representation theory tools

6.1 Finite-representation-type or not?

Gabriel 1972: kQ for quiver Q over field k has finitely many indecomposable representations iff Q's underlying graph is a Dynkin diagram (A_n, D_n, E_6, E_7, E_8).

For substrate quiver: Phase A 36-node finite subgraph has ~774 arrows + 468 fiber arrows. Far from Dynkin shape (too many arrows per node). NOT finite-representation-type.

Tame or wild?: - Tame: one-parameter family of indecomposables in each dimension - Wild: parameter families with multiple parameters

Substrate quiver with multiple SU(2)-like spin operators + gauge fibers is likely wild (high arrow density). v0.3+ explicit computation.

6.2 Auslander-Reiten quiver

For wild quivers, the AR-quiver still exists but is more complex. The AR-translation τ would be a substrate-natural symmetry; almost-split sequences would encode minimum-action transitions.

Substrate-mechanism candidate: τ may correspond to Möbius involution σ (acting on SO(2) factor). v0.3+ verification.

6.3 ADE classification + Lie algebra

D_IV^5 involves SO(5,2). The substrate quiver may have natural connection to a Dynkin quiver of type B_2 (= C_2) for SO(5) × SO(2) → but the SO(5,2) non-compactness makes this non-standard.

Substrate-mechanism candidate: substrate quiver may be non-Dynkin extension of B_2 with additional arrows from non-compact factor. v0.3+ multi-week formalization.

6.4 Derived category

D^b(rep(Q, R)) provides higher-categorical setting. BGG resolutions + Verma module structure may apply per Wallach 1976.

Substrate-mechanism candidate: A_sub operations may have natural homological-algebra interpretation; the 10 closed commutators may be encoded as exact triangles. v0.3+ multi-month formalization.

7. Cal Thread 4 typing request

Per Cal #122 type-A geometric / type-B algebraic / type-C level-crossing:

Multi-phase quiver framework type:

- Type A (geometric): K-type nodes are geometric (Wallach lattice on D_IV^5); arrows are derived from algebra acting on geometry
- Type B (algebraic): A_sub commutators are algebraic; quiver is algebraic encoding
- Type C (level-crossing): bridging Wallach K-type geometry with A_sub algebra at the quiver level

My prior (Cal #27 STANDING applied): Type C level-crossing. The K-type nodes are geometric content (Wallach K-type lattice points on D_IV^5); the A_sub-induced arrows are algebraic content (commutator relations); the path algebra $kQ/\langle R\rangle$ identifies the two at the quiver-representation level.

Multi-phase structure: each sub-quiver may have different type: - Q_DIRECT: predominantly Type A (geometric — Shilov boundary projection) - Q_COMPOSITION: predominantly Type B (algebraic — RG-flow algebra) - Q_MATERIAL: Type C (level-crossing — geometry × external BC) - Q_COMBINATORIAL: Type B (algebraic — number-relations)

If this typing holds: substrate framework has organized type structure at each operational phase region.

8. Substantive consequences

8.1 A_sub algebra is now explicitly a quiver representation

With the kQ framework: A_sub's 10 closed commutator algebra IS the relation ideal R of a multi-phase quiver Q with 36 K-type nodes + ~774 main arrows + 468 fiber arrows. The substrate's Bergman $H^2(D\_IV^5)$ is the canonical representation.

This brings the BST substrate into well-studied mathematical machinery (Gabriel's theorem, AR-quiver, derived categories) with substrate-specific instance.

8.2 Multi-phase quiver candidate strengthened

v0.6 candidate (multi-phase quiver as math-object answer to Casey's standing meta-question) gains explicit framework + computability: - Explicit quiver definition (Section 2) - Explicit path algebra framework (Section 3) - Explicit relation ideal (Section 3.4) - Phase-specific sub-quiver decomposition (Section 5) - Representation theory tool applicability framework (Section 6)

v0.3+ multi-week computation work: - Reduce kQ_Phase A / ⟨R⟩ explicitly via Gröbner basis or similar - Compute indecomposable representations - Identify Dynkin / non-Dynkin extension type - Apply AR-quiver structure - Connect to derived category

8.3 Bridge to Lie algebra structure

The substrate quiver may have natural Lie algebra structure via Hall algebra construction (Ringel 1990). The A_sub generators correspond to quiver arrows; commutator relations correspond to path relations; the Lie algebra hopf structure follows.

Substrate-mechanism candidate: A_sub's Lie algebra structure recovered as Hall algebra of the substrate quiver. v0.3+ multi-month formalization.

9. Honest scope (Cal #27 STANDING)

What's STRUCTURALLY VERIFIED in v0.8: - Standard quiver representation theory applies (Gabriel 1972, etc.) - Path algebra kQ construction is standard - Relations R from A_sub commutators (Cal #132 SVC for 8 + step 10 SVC CANDIDATE + step 9 FRAMEWORK-PLUS) - 36-node Phase A v0.2 set is finite

What's FRAMEWORK in v0.8: - Multi-phase quiver as the math-object identity for K-type graph (Section 2-5) - 4 sub-quiver decomposition with phase-specific relations (Section 5) - Type-C level-crossing typing prior (Section 7) - Representation theory tool applicability (Section 6)

What's INTERPRETIVE in v0.8: - Substrate quiver as Hall algebra base for A_sub Lie algebra (Section 8.3) - Substrate AR-translation τ as Möbius involution σ (Section 6.2) - Non-Dynkin extension of B_2 conjecture (Section 6.3)

What's NOT in v0.8 (multi-week+): - Explicit kQ / ⟨R⟩ quotient computation (v0.3+) - Indecomposable representation classification (v0.4+) - Dynkin type identification or non-Dynkin proof (multi-week) - AR-quiver structure explicit (multi-week) - Derived category structure (multi-month) - Hall algebra connection to A_sub Lie algebra (multi-month) - Cal Thread 4 type-check resolution (own-cadence)

Cal #27 STANDING reflexive trigger count: 2 triggers (Section 6.2 AR-translation as Möbius involution; Section 8.3 Hall algebra connection). Both flagged INTERPRETIVE; require multi-week verification.

Cal #29 STANDING applied at design: question-shape audit passed at Section 1. Framework derivation is structural; not back-fittable.

10. v0.3+ multi-week computation plan

1. Reduce kQ_Phase A / ⟨R⟩ explicitly: apply 10 commutator relations to compute reduced path algebra basis. Mechanical computation; multi-week. Tools: Gröbner basis algorithms, computer algebra.
2. Indecomposable representations: enumerate indecomposable kQ/⟨R⟩-modules. Multi-week per phase region.
3. Dynkin/non-Dynkin classification: determine whether substrate quiver underlying graph is Dynkin (finite-rep-type), tame (one-parameter families), or wild (multi-parameter). Multi-week.
4. AR-quiver explicit: compute AR-quiver structure; identify τ-translation; almost-split sequences. Multi-month.
5. Hall algebra connection: build Hall algebra Hall(kQ/⟨R⟩) and compare to A_sub Lie algebra. Multi-month.
6. Cal Thread 4 type-check: resolve Type A/B/C per sub-quiver; promote v0.8 framework based on typing.
7. Phase B extension: extend to ≤8 (45 nodes) or ≤10 (66 nodes) cutoff per Elie Toy 3536 sizing. Substantially more arrows; multi-month per cutoff.

11. Coordination

Cal: Thread 4 cold-read on Section 7 type-check + Section 6 representation theory tools applicability + Section 8.3 Hall algebra connection candidate. Multi-week priority Type-C level-crossing typing.

Elie: Toy 3538+ candidate — Phase B extension to ≤8 cutoff (45 nodes) for downstream quiver computation. Or: Cal #29 pre-audit for Dirac ground state forward derivation (Track P). Multi-day to multi-week.

Grace: catalog cross-references for substrate quiver Section 6 representation theory tools + Section 8 Hall algebra connection. Multi-week own-cadence.

Keeper: integration into Vol 15 Ch 9 case study draft — multi-phase quiver candidate is potentially load-bearing math-object identity for K-type graph.

— Lyra, Multi-phase quiver v0.8 explicit kQ framework filed Tuesday 2026-05-26 ~10:30 EDT per Casey all-tracks authorization + Keeper Task #358 + Decision B sequence after step 10. FRAMEWORK grade with substantive math-object formalization. Substrate quiver = (36 K-type vertices + ~774 main arrows + 468 fiber arrows) with 10-commutator relation ideal R; canonical representation IS Bergman $H^2(D\_IV^5)$. Multi-phase decomposition Q = Q_DIRECT ⊕ Q_COMPOSITION ⊕ Q_MATERIAL ⊕ Q_COMBINATORIAL; phase-specific relations encode substrate operational regions. Multi-week v0.3+ computation path identified. Cal Thread 4 type-check queued for Type-C level-crossing prior.